

OFDG: a laptop-sized numerical preview

Initial evidence for choosing the next larger experiments

This brief compares accuracy, shock resolution, and positivity on small paper-inspired tests. It supports a supervisor discussion; it does not establish a universal method ranking.

What the completed runs suggest

Smooth accuracy: on the finest P2 grids, KXRCF gating keeps errors close to DG; ungated OFDG has larger errors.

Scalar shocks: filtering suppresses the large sampled DG excursions. The two OFDG variants are often nearly coincident.

Next decision: use Shu-Osher wave preservation and curved geometry to distinguish the methods more carefully before scaling up.

Experiment	Configuration	Question
Smooth advection	P1-P3; N=32/64/128/256; T=1.1	Accuracy across order
Piecewise advection	P2; N=128/256; T=1.1	Oscillations and smearing
Burgers	P2; smooth N=32/64/128, T=0.6; shock N=128/256, T=2.2	Nonlinear shock formation
Lax	P1/192, P2/128, P3/96; T=1.3	Equal unknown count
Shu-Osher	P2; N=200/400; T=1.8	Shock-wave interaction
Straight / curved transport	P2; 8 ² /16 ² /32 ² cells; T=1	Geometry and accuracy
2D Riemann	P2; 32 ² /64 ² cells; T=0.25	Wave structure and positivity

Methods and notation

DG is discontinuous Galerkin without the oscillation filter. **OFDG** is its oscillation-free variant; **OEDG** is the oscillation-eliminating comparator. **KXRCF gating** applies OFDG only in cells flagged by a troubled-cell indicator. Pk denotes polynomial degree k; N is the 1D cell count. Squared counts specify 2D grids.

All runs use RK4. CFL is 0.15 for scalar problems and 0.30 for Euler. OFDG filters after each full step; OEDG filters at RK stages. Every Euler method, including DG, uses the same positivity limiter. OEDG appears on selected finer affine shock cases only.

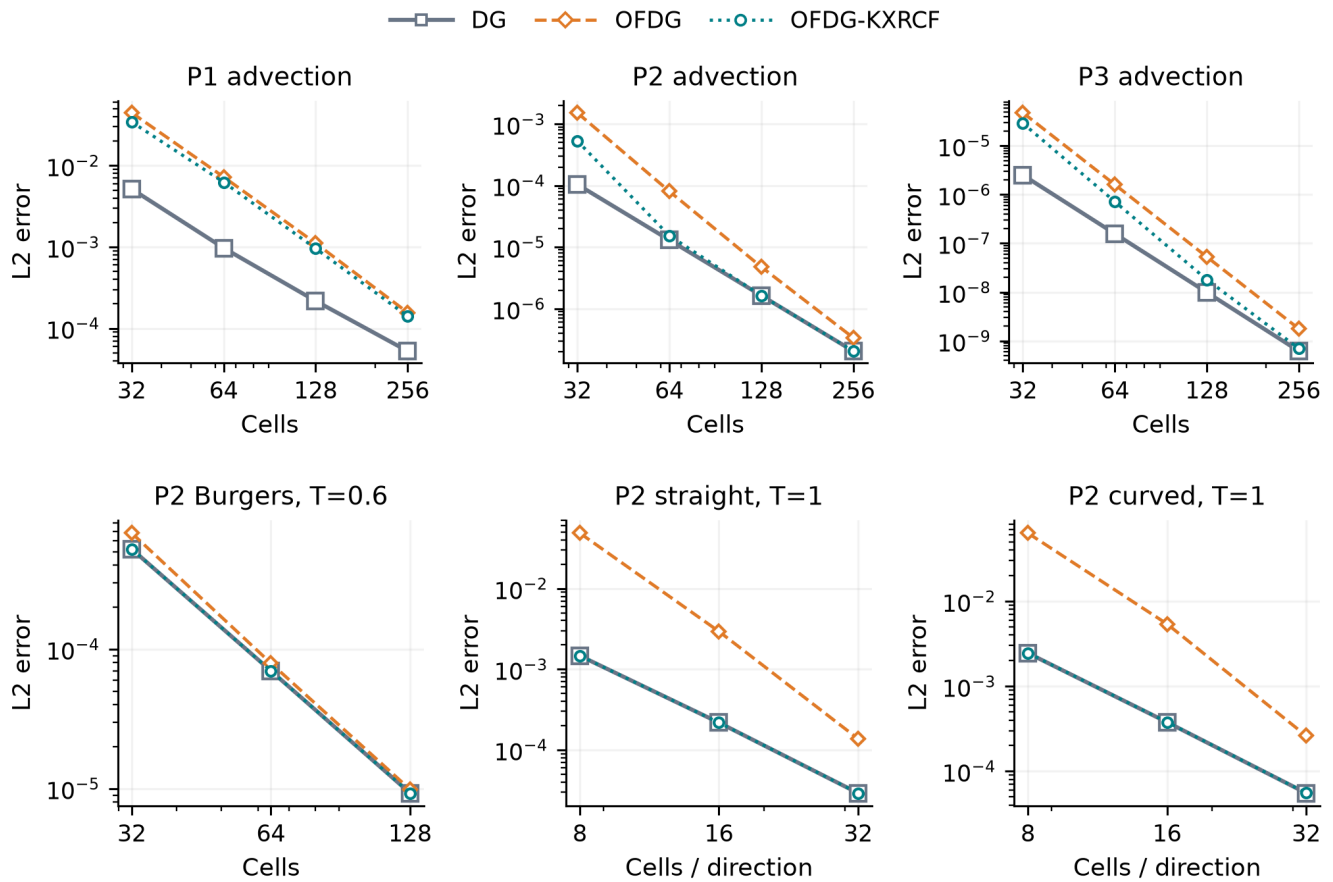
Compute used

103 solver runs and 4 reference solves completed in 7.35 minutes, with sampled peak process-tree memory 118 MiB. Runs were sequential and used at most two MPI ranks. No cases failed or were dropped.

Ceilings: 4 GB, five minutes per check, 60 minutes total; no new checks after 45 minutes. Memory peaks are sampled. Build and document preparation are excluded from solver time. Single-run timings do not establish performance differences.

Smooth accuracy and curved geometry

Refinement reduces every smooth error. Higher slopes in the filtered cases on these grids do not establish a higher asymptotic order.



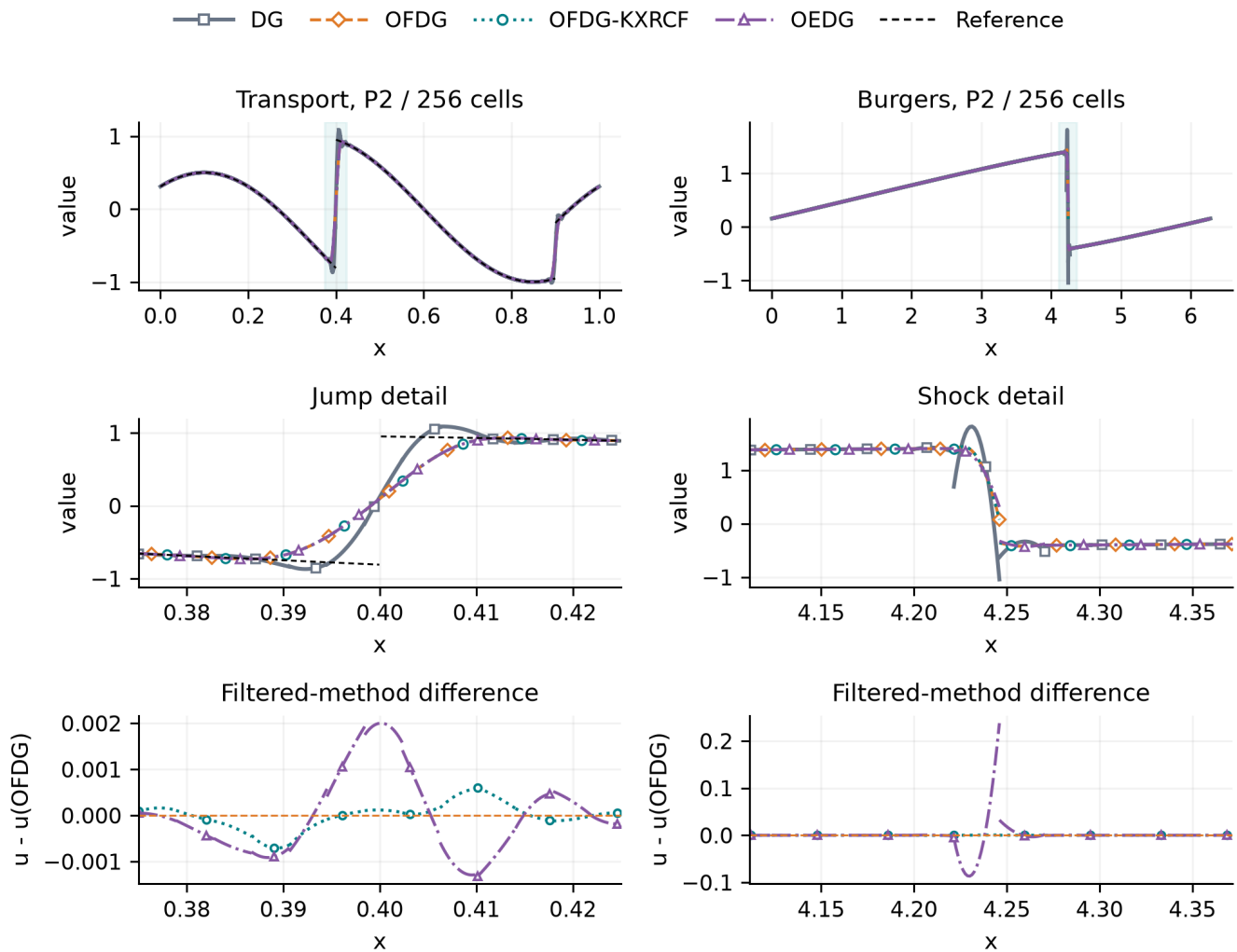
Method	P1 slope	P2 slope	P3 slope	Curved slope	P2 error / DG	Curved error / DG
DG	2.05	3.00	4.00	2.77	1.000	1.000
OFDG	2.83	3.83	4.90	4.37	1.648	4.715
OFDG-KXRCF	2.77	3.00	4.70	2.77	1.000	1.002

Slopes use the finest two grids. Nominal spatial orders are 2, 3, and 4 for P1, P2, and P3. Error ratios use $N=256$ (1D P2) and 32^2 (curved P2). Nested markers indicate nearly coincident curves: the table quantifies the hidden DG / gated-OFDG difference.

The straight and curved 2D panels share the unit-square transport problem: velocity $(0.7, 0.3)$, $\sin^2(\pi(x+y))$ data, and a static cubic deformation for the curved case. This is a 2021-paper-inspired adaptation.

Finest-P3 CFL halving changed L2 error by DG 0.0006%, OFDG 0.0029%, OFDG-KXRCF 0.0003%. No change exceeded 5%, so no extra series was needed. Maximum reported periodic scalar conservation drift: $4e-14$.

Scalar shocks: separate near-coincident curves



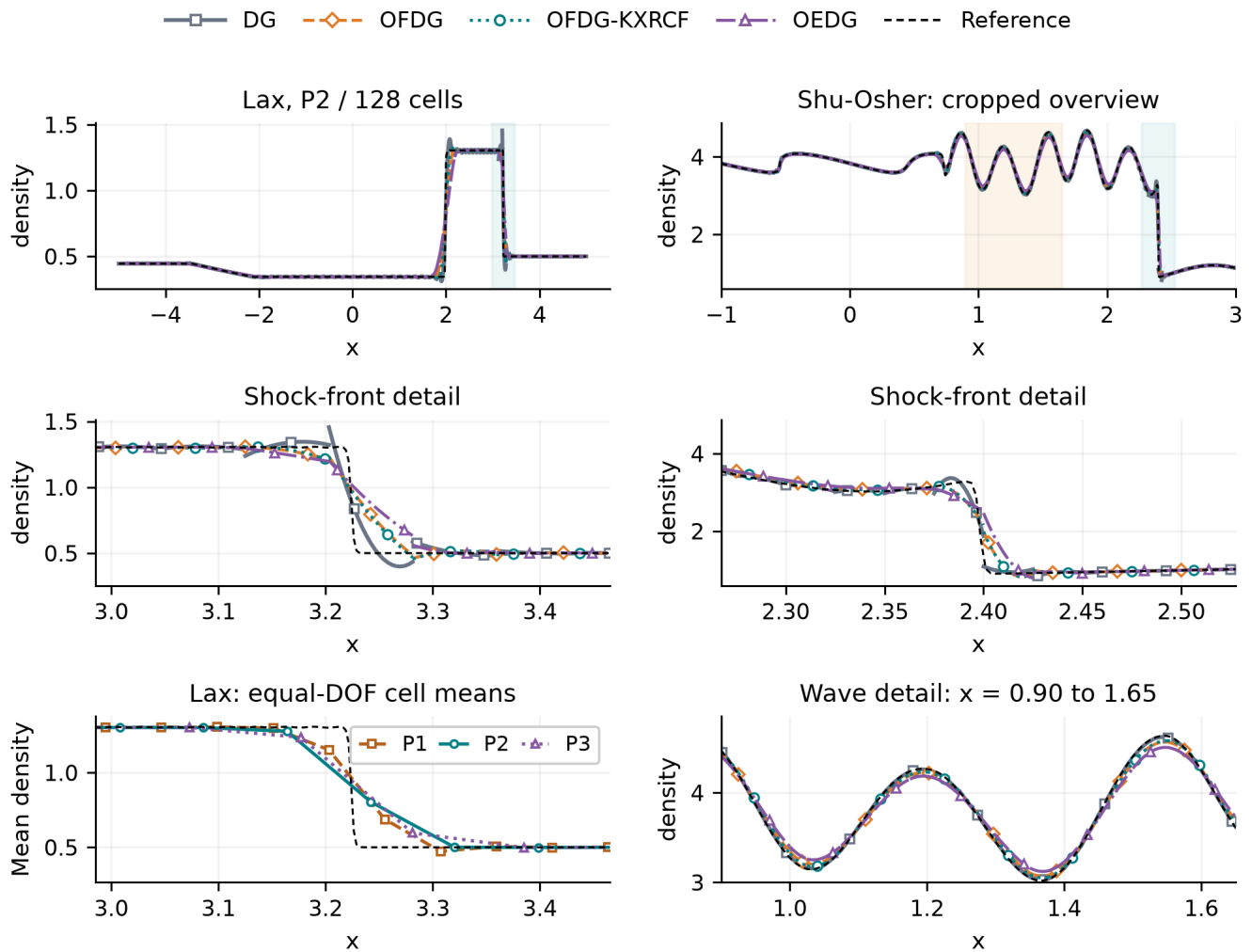
Overviews locate the shaded close-ups. The bottom row subtracts OFDG, so small differences between the filtered methods remain visible. The black reference is exact translated transport; Burgers has no reference curve.

Sampled excursion	Transport under	Transport over	Burgers under	Burgers over
DG	0.012	0.136	0.541	0.317
ODFG	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$
ODFG-KXRCF	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$
OEDG	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$

Excursions are beyond initial-data bounds: transport $[-1, 0.951057]$, Burgers $[-0.5, 1.5]$. Values below 10^{-6} are grouped for display; full precision is retained in analysis.json. The plots use 21 samples per cell, with curves broken at DG interfaces; sampled extrema are not certified continuous extrema.

In the displayed windows, $\max |\text{gated OFDG} - \text{ODFG}|$ is 0.00071 for transport and 0.00049 for Burgers. Near coincidence is a result here, not evidence that the curves were omitted.

Euler: shock fronts and wave detail



Blue bands mark shock details; the orange band marks the Shu-Osher wave detail. Shu-Osher uses P2 / 400 cells; its overview is cropped to $[-1, 3]$ from the $[-5, 5]$ domain. The equal-DOF Lax panel uses gated OFDG with P1/192, P2/128, P3/96: 384 scalar unknowns each. Its markers are cell means, not polynomial samples.

Wave-window peak-to-trough density range: DG 1.590, OFDG 1.495, OFDG-KXRCF 1.529, OEDG 1.383, WENO 1.615. This local range is a first look at amplitude; it does not measure phase accuracy.

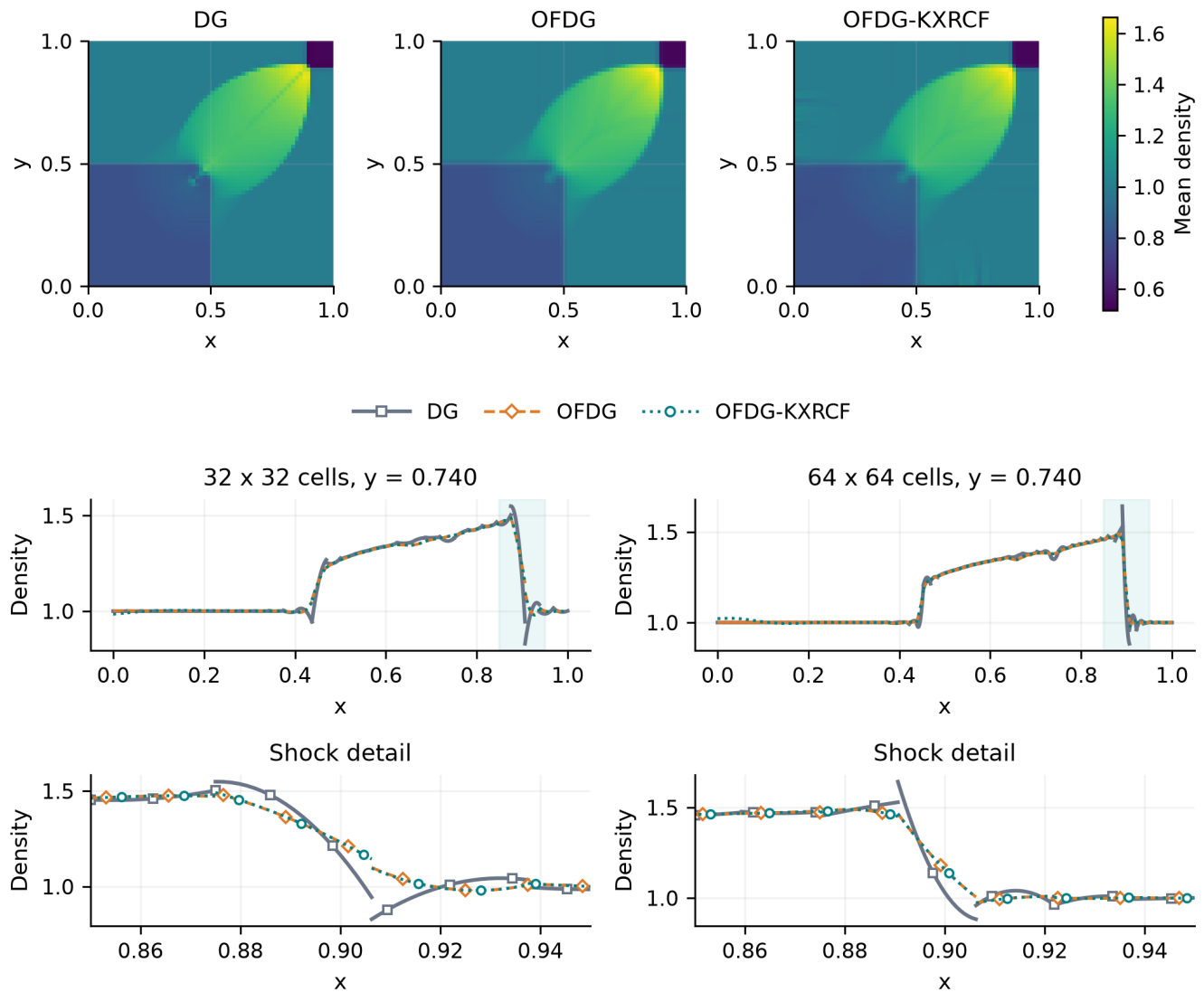
WENO reference pair	Mean absolute difference	Maximum difference
Lax 2048 vs 4096	0.00101	0.12
Shu-Osher 2048 vs 4096	0.00324	0.463

These are density differences after averaging the finer WENO reference onto the coarser grid. Large localized differences near steep fronts limit pointwise judgments even when mean differences are small. WENO5 / SSPRK3 is independent of DG, but is not exact.

Shu-Osher positivity cell-limit events: DG 6252, OFDG 3, OFDG-KXRCF 3, OEDG 2. These are cumulative interventions, not distinct cells. Fixed far-field boundaries permit changes in domain totals; those changes are not conservation-error estimates.

2D Riemann: compare the same physical cut

P2 Euler, $T=0.25$; four-quadrant data with outflow boundaries. Maps show 64^2 physical cell means on one shared scale. Its range spans all displayed means, so no values are clipped.



Both grids are evaluated at exactly $y=0.740$ by reconstructing each P2 polynomial from its tensor-node samples. Curves remain broken at cell interfaces. The two resolutions are separated to keep method comparisons readable.

The two OFDG traces nearly coincide. Max $|\text{gated OFDG} - \text{ODFG}|$ along this cut: 0.0173 on 32^2 and 0.0226 on 64^2 . The close-ups use polynomial traces; the maps above use cell means.

Method / cells	Min. density	Min. pressure	Limit events	Rejected steps
DG / 32^2	0.38	0.207	122	0
ODFG / 32^2	0.513	0.379	0	0
ODFG-KXRCF / 32^2	0.513	0.379	0	0
DG / 64^2	0.431	0.236	482	0
ODFG / 64^2	0.515	0.384	0	0
ODFG-KXRCF / 64^2	0.515	0.384	0	0

Minima are the solver-reported final values. Limit events count cumulative cell interventions. All three methods stayed admissible with the common positivity treatment; these coarse grids do not establish a resolution-independent ranking.

Decisions for the next compute allocation

Priority	Evidence from this preview	Next experiment / decision
1	Shu-Osher filters look close; amplitude alone cannot establish phase or wave accuracy.	Refine the wave region; measure amplitude and phase against a checked reference. Repeat timings only for this selected case.
2	Gating nearly matches DG smooth errors; curved errors are larger on the same grid counts.	Refine matched straight / curved problems, then a small curved Euler case. Separate geometry error from filtering.
3	2D OFDG variants differ little on the sampled cut, while DG uses more positivity interventions.	Refine the Riemann problem with boundary conditions and limiter settings matched across future comparisons.
Later	No result here requires a giant test to interpret.	Defer blast waves, 3D, and scaling studies until the scientific question and hardware are chosen.

What remains uncertain

This is a small configured comparison, not a full paper reproduction. Fluxes, initialization, filtering cadence, and positivity treatment affect the outcome. Only smooth P3 advection has a time-step sensitivity check. Shock widths and the wave-detail range need broader refinement. Native NURBS is outside the preview; curved results use supported polynomial geometry.

Open the evidence

Run: **preview**, 103 successful solver runs and four references.
Dataset: **measurements/study/preview/**
Instructions: **docs/supervisor-preview.md**
Configuration: **experiments/preview_manifest.json**

Paths are relative to `codex_dev`. The dataset retains exact commands, logs, physical samples, reference CSVs, convergence tables, and source/executable fingerprints. Existing thesis assets are unchanged. Project revision `f50a6dff8f98`; MFEM `a1ce49fb5742`.

Research basis

[1] J. Lu, Y. Liu, C.-W. Shu (2021). An Oscillation-Free Discontinuous Galerkin Method for Scalar Hyperbolic Conservation Laws. Scalar advection and Burgers examples. DOI: [10.1137/20M1354192](https://doi.org/10.1137/20M1354192)

[2] Y. Liu, J. Lu, C.-W. Shu (2022). An Essentially Oscillation-Free Discontinuous Galerkin Method for Hyperbolic Systems. Lax, Shu-Osher, and equal-DOF comparisons. DOI: [10.1137/21M140835X](https://doi.org/10.1137/21M140835X)

[3] M. Peng, Z. Sun, K. Wu (online 2024; issue 2025). OEDG: Oscillation-Eliminating Discontinuous Galerkin Method for Hyperbolic Conservation Laws. First four-quadrant Riemann setup. DOI: [10.1090/mcom/3998](https://doi.org/10.1090/mcom/3998)

Curved derivatives: does order matter?

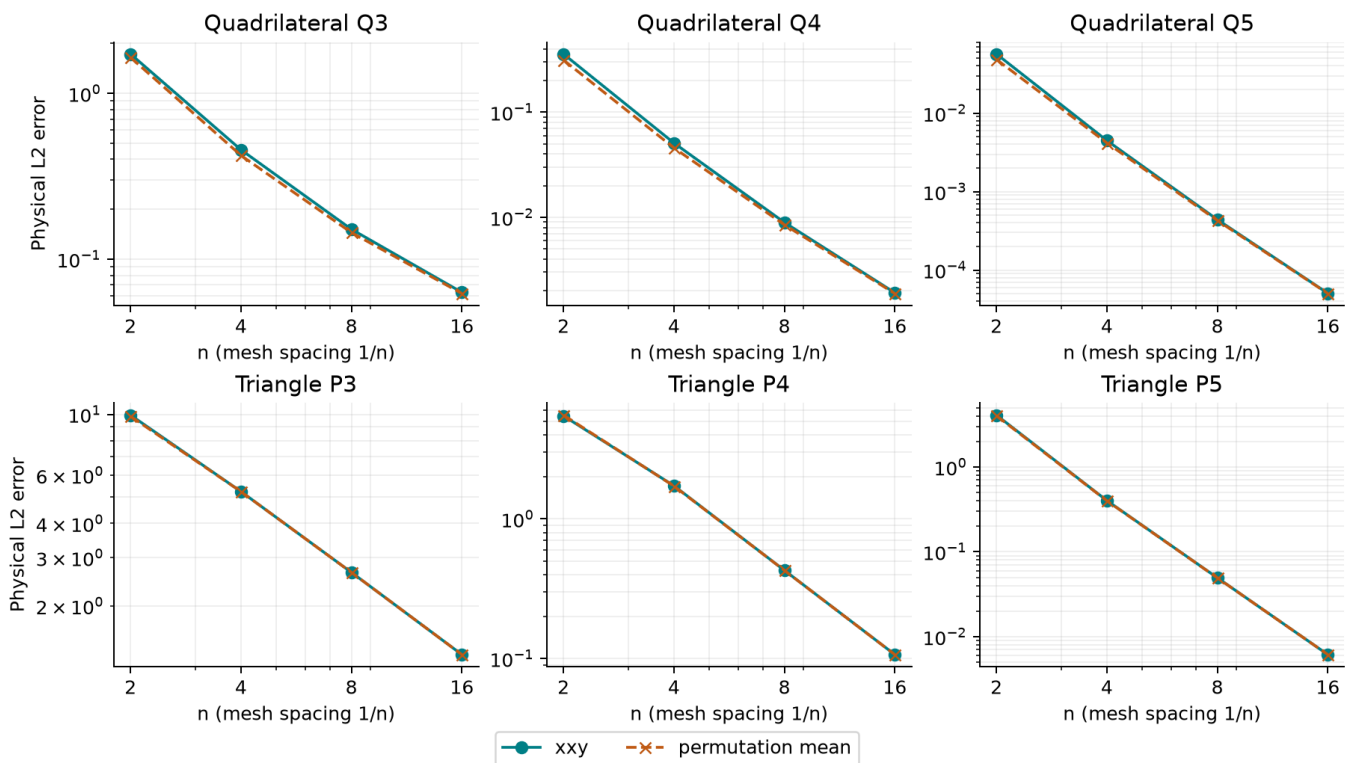
Isolated manufactured-derivative experiment; no PDE evolution and no change to the production OFDG algorithm. Exact mixed partial derivatives commute. This check measures the effect of inserting a physical L2 projection after each differentiation.

Construction and comparisons

Domain and geometry: start from (s,t) in $[0,1]^2$; set $b = a s(1-s)t(1-t)$, $x = s+b$ and $y = t+0.7b$. Use exact degree-4 polynomial geometry with $a = 0, 0.4, 1.2$ (affine, mild and stronger curvature). The domain boundary stays fixed; $\det J$ is at least $1-0.425a > 0$.

Spaces and data: triangle P3-P5 and quadrilateral Q3-Q5; $n = 2, 4, 8$, plus $n = 16$ at $a = 1.2$. Triangles have $2n^2$ elements, quadrilaterals n^2 ; these are not equal-DOF comparisons. Physically L2-project $\sin(2x+3y)$, $\exp(x+y/2)$, and x^2y+xy^2 into each space before differentiating.

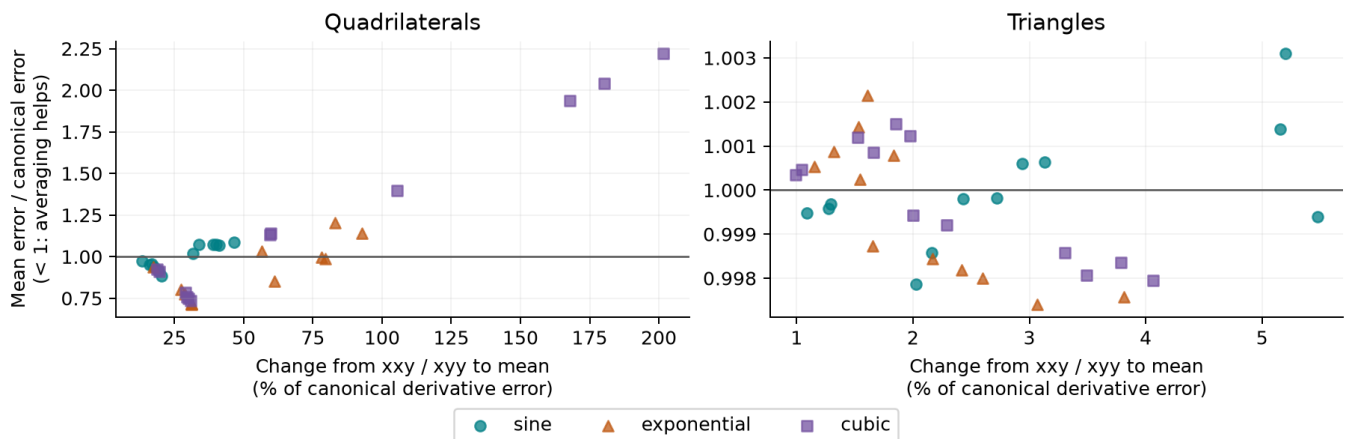
Sequences: words denote chronological operations: xy means x , then x , then y . Test xy/yx ; $xy/xyx/yxx$; and $xy/yxy/yxx$. The canonical sequence matches the production derivative graph. Average all distinct permutations as functions, then measure physical L2 errors against the analytic derivative. The average is a candidate approximation, not a reference solution.



Refinement: sine field, $a = 1.2$, xyx derivative. The exact derivative is $-12 \cos(2x+3y)$. Solid circles show the production order; dashed crosses show the permutation mean. Near-overlap is real; the next page plots changes directly. Both approaches improve under refinement in all six configurations.

Errors include differentiation of the initial solution projection error as well as the intermediate derivative projections. A separate direct projection of the analytic derivative supplies an approximation floor in the raw data. This is not a comparison with the exact higher derivative of the represented DG function.

Averaging: a change, not a guaranteed gain



Each point is one curved $n=8$ case: P/Q3-P/Q5, both nonzero curvature amplitudes, and xy or xyy . Horizontal position measures the L2 change to the derivative, divided by its canonical error. Vertical position measures the resulting error ratio. Points above 1 became less accurate after averaging. Axes use different scales: the triangle accuracy changes are much smaller.

Sine, $a=1.2$, $n=8$	Canonical error	Mean error	Error change
Q3 / xyy	0.1515	0.1446	-4.56%
Q4 / xyy	0.008933	0.008491	-4.95%
Q5 / xyy	0.0004397	0.0004283	-2.59%
P3 / xyy	2.649	2.649	-0.02%
P4 / xyy	0.4284	0.4278	-0.14%
P5 / xyy	0.04932	0.04922	-0.21%

Across 72 curved $n=8$ third-derivative cases, averaging lowers the error in 41. Mean/canonical error ratios range from 0.712 to 2.220 (median 0.999). The derivative change itself ranges from 1.0% to 201.6% of canonical error.

The largest worsening is mild-curvature Q3, cubic field, xyy : error 0.00109 to 0.00242 (exact derivative = 2). The relative errors are only 0.055% and 0.121%; the factor alone exaggerates its absolute significance.

Controls and interpretation

Affine canonical-to-mean differences are at most $5.6e-09$ in L2; the affine cubic derivative error is at most $3.3e-09$. Independent first-derivative weak residuals are below $3.4e-15$; constant-derivative coefficients below $8.5e-12$. Raising assembly quadrature by eight on four strong-curvature cases changes reported error norms by at most $2.4e-05$ relative.

Decision: retain the canonical implementation for now. Averaging removes the arbitrary ordering convention, but does not reliably reduce derivative error. These volume norms do not measure OFDG face-jump rates, damping, or final PDE accuracy. A targeted face-sensor and smooth-advection comparison is the next step before changing the method.

Resources and evidence: 64 sequential checks, one rank/thread; 4.63 minutes total, sampled peak 19.9 MiB. No new large PDE runs. Raw data, commands, hashes, quadrature controls and derived tables: [measurements/derivative-permutations/](#). Reproduction: [docs/derivative-permutations.md](#). This addendum is separate from the thesis.

How accurate are the derivatives?

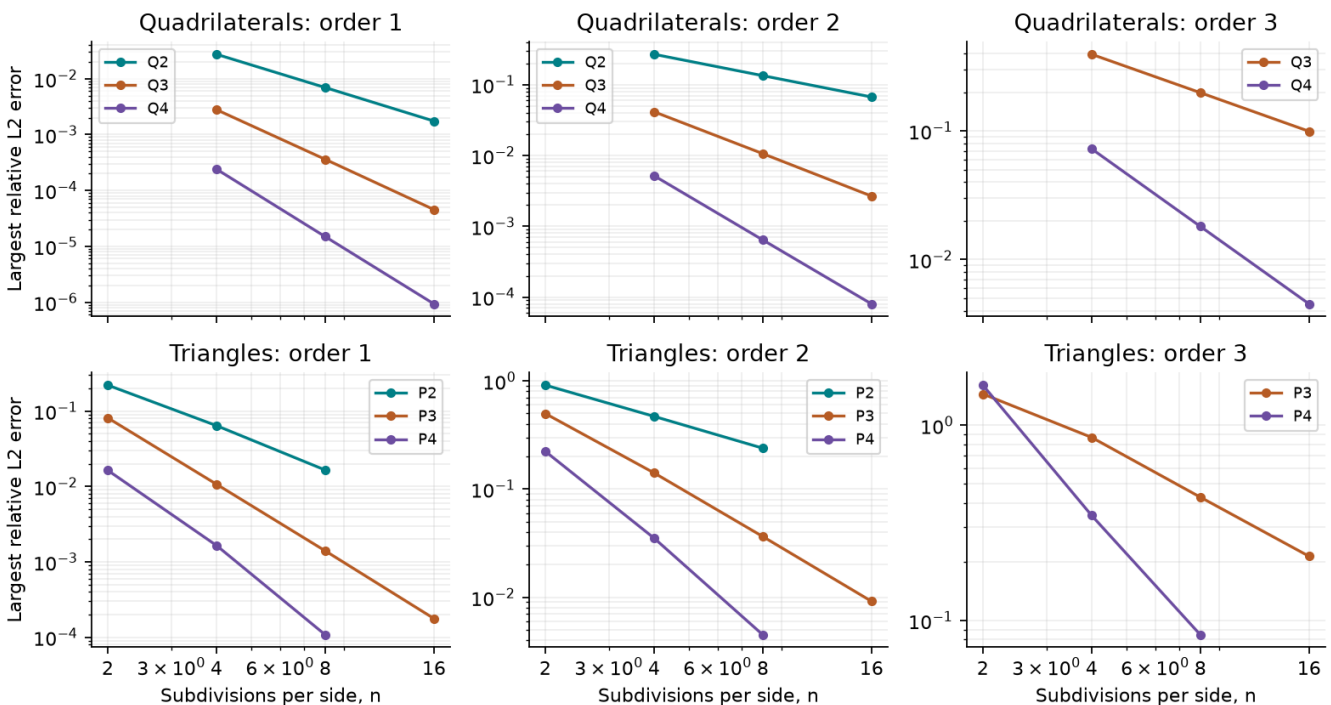
An isolated operator study compares the production projected-derivative matrices with analytic physical derivatives. No PDE is advanced and no production filtering code is changed. Higher derivative errors can be substantial even when first derivatives are accurate.

Two questions, tested separately

A. Complete approximation: physically project $\sin(2x+3y)$, $\sin(8x+6y)$, and $\exp(x+y/2)$, then differentiate recursively. Errors include the input projection and all intermediate derivative projections. Use the coupled curved-face map from page 7, with $a=0$, 0.4, 1.2.

B. Derivative projection alone: use an exactly representable mapped quadratic and an analytically invertible non-affine map (next page). This removes input approximation error and permits comparison with exact derivatives of the represented DG function itself.

Coverage: P/Q1-P/Q4 for A; P/Q2-P/Q4 for B. All canonical coordinate derivatives through $\min(p,3)$: $x, y, xx, xy, yy, xxx, xxy, xyy, yyy$. Gauss-Lobatto, Gauss-Legendre and positive Bernstein bases. Refinement uses $n=4,8,16$ for quads and $n=2,4,8$ for triangles, with selected $n=16$ triangle checks. Basis comparisons use $n=8$; the raw manifest identifies every case. Geometry degree 4 represents both maps exactly. Spaces at equal n are not equal-DOF comparisons.



Complete-approximation errors: $\sin(2x+3y)$, strong coupled curvature $a=1.2$, Gauss-Lobatto basis. Each curve shows the largest relative volume L2 error among derivatives of the indicated order. Different panel scales expose the refinement trend; high derivative orders remain much less accurate. These are canonical sequences, not permutation averages.

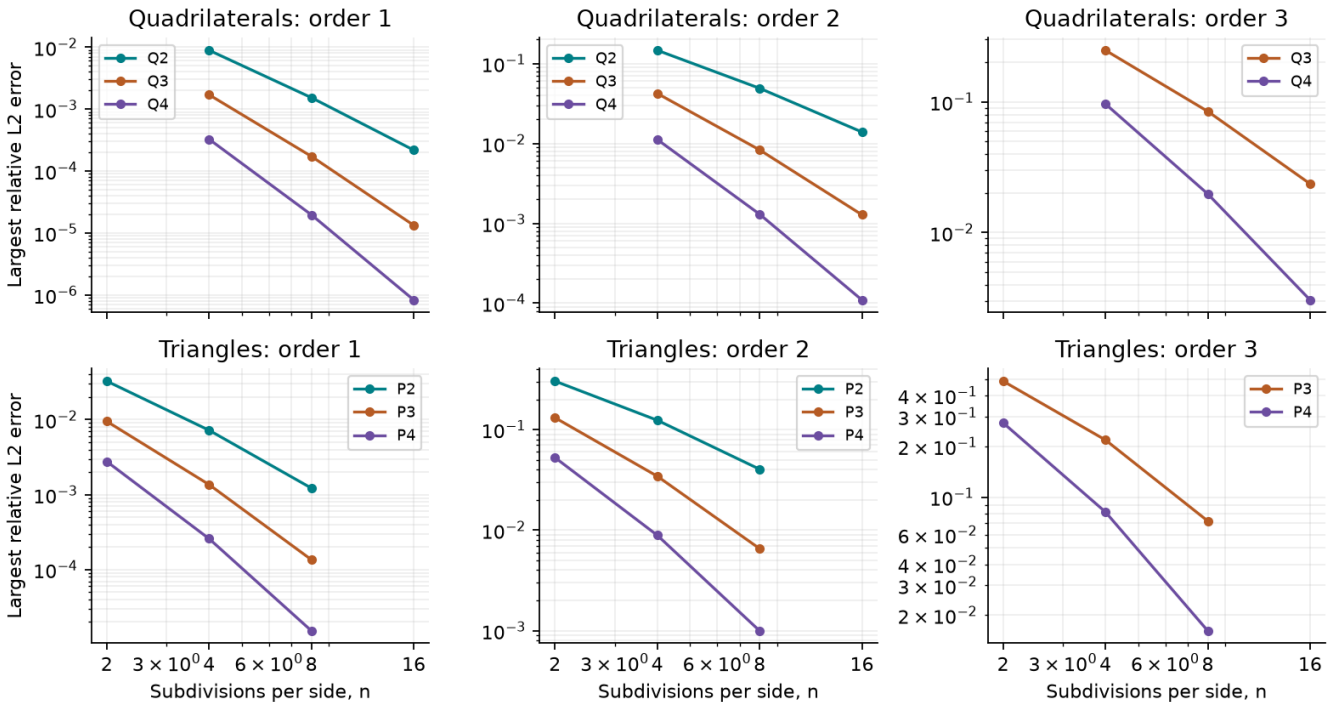
What “orders of magnitude off” means here

Absolute error $E = \|D^{\alpha} u_h - \text{partial}^{\alpha} u\|_{L_2}$; relative error $R = E / \|\text{partial}^{\alpha} u\|_{L_2}$. Thus $R=10^{-2}$ means a 1% norm error, two orders below the derivative norm. It is not a pointwise guarantee of two correct digits. Zero analytic derivatives have undefined relative error and are reported with absolute errors. No regularizing denominator hides those cases.

Separate projection error from input error

For the isolation test, take $x=s+a s(1-s)$, $y=t+0.7a t(1-t)$, and $u(x,y)=s(x)^2+s(x)t(y)+t(y)^2$. Test $a=0.3$ and 0.7 . On every element its pullback is quadratic, so P/Q2 and above represent it exactly. Its physical derivatives are generally nonpolynomial. The inverse map and its first three derivatives are analytic; see the reproducibility note.

The separable map stretches coordinate lines nonlinearly. Quad edges stay straight, while triangle diagonals can curve; it isolates non-affine parameterization rather than reproducing the curved quad faces of study A. Both maps are nonsingular throughout the domain.



Projection-only errors: strong separable map $a=0.7$. Initial physical projection error is at most $1.3e-14$ in L2 across the tested cases. All plotted errors can therefore be attributed to derivative representation, recursion and numerical roundoff, rather than an unresolved input field.

n=8, strong map	First derivative	Second derivative	Third derivative
Q2	3.08e-03 0.153% (x)	9.22e-01 4.91% (xx)	not used
Q3	3.47e-04 0.0173% (x)	1.56e-01 0.831% (xx)	5.25e+01 8.46% (xxx)
Q4	3.95e-05 0.00196% (x)	2.44e-02 0.13% (xx)	1.23e+01 1.98% (xxx)
P2	2.46e-03 0.122% (x)	7.58e-01 4.04% (xx)	not used
P3	2.73e-04 0.0136% (x)	1.23e-01 0.655% (xx)	4.48e+01 7.22% (xxx)
P4	3.08e-05 0.00153% (x)	1.87e-02 0.0997% (xx)	1.00e+01 1.61% (xxx)

Each cell gives absolute L2 error, then relative error; the displayed derivative has the largest relative error at that order. The worst absolute-error direction may differ. A direct physical projection of each exact derivative supplies the best-approximation floor. Higher recursive errors can exceed this floor because earlier projection errors are differentiated again. The raw results include both the floor and the excess over it.

Implications for OFDG and checks

How large are complete-approximation errors?

n=8; a=1.2	Order 1	Order 2	Order 3	Worst face error
Q2 / sine	1.48e-02 0.695%	3.78e-01 13.4%	not used	18.6%
Q3 / sine	7.67e-04 0.0361%	6.73e-02 1.06%	1.12e+00 19.8%	27.6%
Q4 / sine	3.17e-05 0.00149%	4.07e-03 0.0641%	1.03e-01 1.81%	3.2%
P2 / sine	2.34e-02 1.65%	6.73e-01 23.9%	not used	31%
P3 / sine	1.98e-03 0.14%	1.03e-01 3.64%	2.43e+00 42.8%	56.1%
P4 / sine	1.53e-04 0.0108%	1.27e-02 0.45%	4.80e-01 8.46%	13.8%

Volume cells show absolute L2 error / largest relative error over the canonical derivatives at each order. The last column is the largest relative one-sided face L2 error over all tested derivative orders, normalized by the analytic trace norm. It uses every element boundary, counting interior faces twice. It is a trace diagnostic, not the OFDG jump or damping rate.

Across these n=8 P/Q2-P/Q4 strong-curvature cases, the order-wise worst relative errors range from 2.6e-07 to 1.9 for the sine/exponential fields, and 0.00049 to 0.57 for the shorter wave. These are envelopes of a configured test set, not a statistical statement about typical meshes. Sampled maximum errors and nominal-h-scaled errors are also saved.

Basis, integration and analytic controls

Changing among Gauss-Lobatto, Gauss-Legendre and Bernstein bases changes the reported L2 error norms by at most 5e-08 absolute (8.4e-10 on nonconstant fields after scaling by the larger analytic derivative/input norm). This compares error norms, not pointwise equality of the derivative functions. Independent first-derivative weak residuals are at most 3.9e-15. Constant-field sampled derivative magnitudes reach 1.8e-06: repeated differentiation amplifies tiny initialization/assembly roundoff. Relative errors for these zero derivatives are undefined.

Raising assembly quadrature by eight changes error norms by at most 1.8e-10 on that scale; independently raising initialization/evaluation quadrature by eight changes them by 3.2e-10. Evaluation rules otherwise use production order +16. Analytic inverse-map derivatives are checked independently by high-precision finite differences. Sampled maxima are not certified continuous suprema.

Assessment: useful, with a resolution-dependent boundary

Retain this construction for OFDG, but do not describe its higher derivatives as uniformly accurate. Refinement and increased order improve the tested errors; changing coefficient basis does not cure them. First derivatives can be several orders below the true derivative norm while the highest derivative used by a low-order space remains at percent-to-tens-of-percent error. Strong distortion and shorter waves require particular care.

ODFG uses derivatives as roughness indicators, not as a replacement for the DG flux. That makes this a sensible construction to continue evaluating, but volume accuracy alone cannot establish acceptable damping or PDE accuracy. Face errors can be larger. Before claiming the extension is sufficiently accurate, compare face-jump rates and smooth-advection convergence against an independent derivative construction; shock robustness requires a separate check.

Resources: 135 completed checks, 0 failures; one rank/thread on an 8-logical-processor laptop. 14.69 minutes charged compute (including a conservative interruption allowance), peak sampled RSS 21.3 MiB. Raw data, exact commands, controls, source hashes and tables: [measurements/derivative-accuracy/](#). Instructions: [docs/derivative-accuracy.md](#). No new shock/PDE runs; no claim of stability or a universal error bound.

Differentiate the original mapped DG function

Question: can we evaluate physical derivatives of the existing DG function directly, avoiding every intermediate projection? This is an isolated operator experiment. The production OFDG implementation and all earlier raw datasets remain unchanged.

What changes, and what stays the same?

Current recursive projection	Direct mapped differentiation
After each physical derivative, project back into the original mapped DG space. Repeat on the projected coefficients.	Differentiate the original mapped function through the curved coordinate map. Evaluate the requested derivative directly at quadrature points.
Intermediate projections can change higher derivatives and make different derivative orderings disagree.	No derivative projection or new solution space. Mixed partials commute mathematically within each smooth, invertibly mapped element.

Both methods receive exactly the same physical L2-initialized DG coefficients. Direct differentiation does **not** reconstruct a polynomial in physical coordinates and does not use analytic solution derivatives to calculate the numerical answer. Analytic derivatives are used only as error references.

Mathematical construction

Write the reference solution as $u\text{-hat}(\xi) = \sum_j c_j \phi\text{-hat}_j(\xi)$, and the physical function as $u_h(x) = u\text{-hat}(F^{-1}(x))$. With $B = J^{-1}$, physical differentiation in direction i is $L_i = \sum_a B_{ai} \partial/\partial \xi_a$. Repeated application must differentiate B as well as the reference polynomial.

Equivalently, at a point x_q form the third-order Taylor expansion of $F^{-1}(x_q + h z)$. Compose the original reference polynomial with that expansion. Its coefficient of $z_x^a z_y^b$, multiplied by $a! b! / h^{a+b}$, gives the physical derivative with a x -differentiations and b y -differentiations. Truncation to degree three retains all derivatives through degree three; this is not a finite-difference approximation.

Implementation and support boundary

The prototype uses truncated bivariate Taylor arithmetic and four formal inverse-map corrections with the base-point Jacobian. Reference polynomials are recovered in a local monomial basis by interpolation, an algebraic change of basis checked against MFEM values and first derivatives. The known polynomial geometry formula is used directly and checked against MFEM point evaluations. No global inverse-map formula is required.

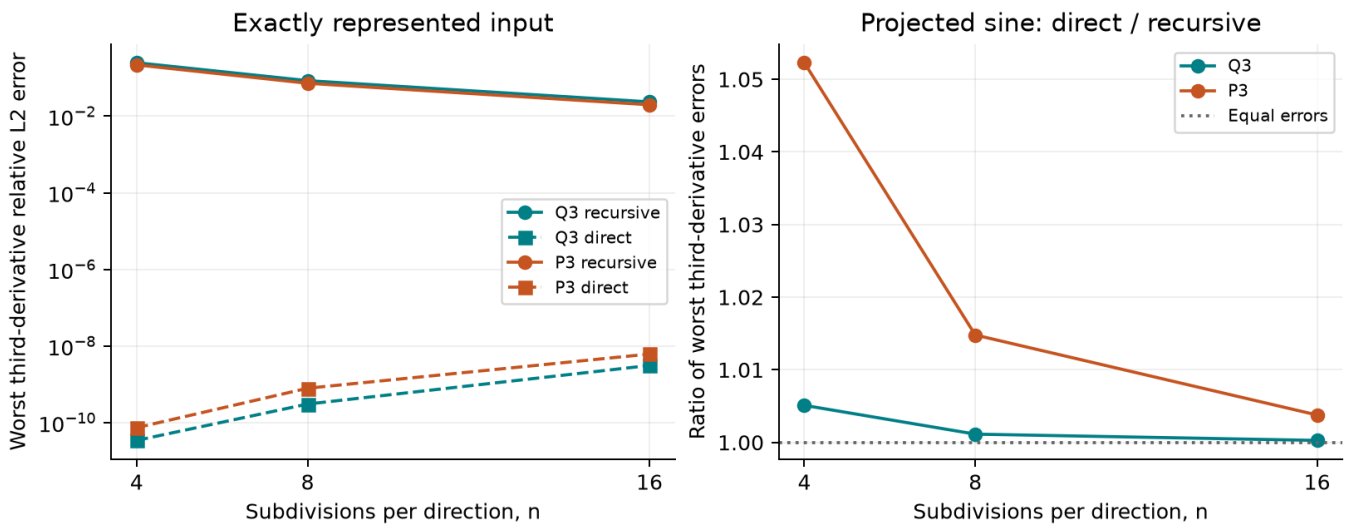
Coverage: polynomial 2D triangles and quadrilaterals, P/Q2-P/Q4, derivatives through $\min(k,3)$, and Gauss-Lobatto, Gauss-Legendre and Bernstein coefficient bases. These tests do not supply a general MFEM higher-geometry-derivative interface, NURBS/3D support, a cached production implementation, or a PDE stability result. Different derivative permutations are implicit in one Taylor coefficient; separate order-by-order implementations were not compared.

What improves, and what does not

Matched-input third derivatives: $n=8$, solution order $k=3$

Input / space	Recursive relative error	Direct relative error	Recursive / direct absolute error
Exact input / Q3	8.464%	1.833e-10%	52.5 / 1.14e-09
Exact input / P3	7.224%	1.376e-09%	44.8 / 8.54e-09
Projected sine / Q3	19.82%	19.84%	1.12 / 1.13
Projected sine / P3	42.83%	43.47%	2.43 / 2.47

Every table entry refers to the same xxx derivative. Relative error is $\| \text{numerical derivative} - \text{analytic derivative} \|_{L_2(K \text{ summed})} / \| \text{analytic derivative} \|_{L_2(K \text{ summed})}$. Absolute errors are physical volume L2 norms. Exact-input third derivatives can be large; percentages and absolute errors therefore answer different questions.



Plots use the largest relative error over xxx, xxy, xyy, yyy; the maximizing direction can differ between methods. Left: direct errors are limited by amplified floating-point/input roundoff, so their increase under refinement is not a truncation-error convergence rate. Right: ratios above one mean direct differentiation has a larger error. Ratios expose differences that would be hidden by overlapping error curves.

Why these outcomes coexist

Direct differentiation removes the recursive operator error relative to the represented u_h . It does not remove the differentiated input error, $\text{partial}^{\text{alpha}}(u_h - u)$. For the exactly represented input, that term vanishes in exact arithmetic and the improvement is dramatic. For the projected sine, differentiation of the existing DG approximation dominates; removing intermediate projections gives little benefit and can slightly increase error. Projection can also suppress components of the input error. No general ranking against the analytic solution follows.

Diagnostic fields and geometry

Original coordinates (s,t) cover the unit square, with n^2 quads or $2n^2$ triangles. Sine: $u = \sin(2x+3y)$, $x=s+b$, $y=t+0.7b$, $b=1.2s(1-s)t(1-t)$. Exact input: $u=s(x)^2+s(x)t(y)+t(y)^2$, $x=s+0.7s(1-s)$, $y=t+0.49t(1-t)$. The latter is a non-affine map with straight quad edges. Both maps are represented exactly by degree-four geometry, up to floating-point error.

Face diagnostics, verification and next decision

Sensor-relevant changes

Strong maps, n=8	Volume error ratio	Face error ratio	Interior jump ratio
Smooth projected fields	1.001	1.002	1
Exactly represented input	5.198e-09	4.058e-09	6.206e-09

Entries are medians of direct/recursive ratios for matched derivative directions: 138 comparisons for sine, shortwave sine and exponential inputs; 46 for the represented quadratic, across P/Q2-P/Q4. Face errors use both one-sided traces; jump RMS uses interior faces, where the analytic jumps vanish. Cases with baseline norms below $1e-9$ are excluded from ratios. These are derivative diagnostics, not full OFDG damping rates.

For the smooth projected inputs, the volume ratios range from 0.983 to 1.373 and the jump ratios from 0.565 to 1.996. Direct differentiation is therefore not uniformly better for those fields. For the exactly represented input, all 46 volume, face and jump comparisons improve. A smaller jump alone does not establish a better shock sensor.

Independent checks and numerical limits

Higher derivatives: 18 first-to-third derivative probes on both maps agree with independently inverted, 65-digit physical finite differences within $1.1e-10$, scaled by $\max(1, \text{reference magnitude})$. Changing the finite-difference step from $1e-5$ to $1e-6$ changes the reference by at most $1e-08$. Finite differences validate the jet method; they are not used by it.

Representation and baseline: polynomial evaluation differs from MFEM by at most $7.3e-14$; physical first derivatives differ by $1.8e-11$ on a $\max(1, \text{magnitude})$ scale. Geometry positions differ by at most $1.4e-13$ after division by h . The 934 matched recursive error norms reproduce the prior accuracy study exactly in the saved CSV values.

Sensitivity: changing input basis changes error norms by at most $1.1e-09$, scaled by $\max(1, \text{analytic derivative L2 norm})$; eight extra evaluation quadrature orders change them by $2e-14$. Direct represented-input relative errors are at most $6.2e-09$ across the suite. Constants have up to $2.1e-08$ absolute derivative L2 error: differentiation amplifies initialization and basis-conversion roundoff. No exact-zero claim is made.

Decision

Retain the existing production method; keep direct differentiation as a promising prototype. It resolves the projection-error mechanism decisively for exactly represented inputs, but does not generally improve derivatives of projected smooth inputs. Before adoption, compare actual OFDG damping rates and PDE behaviour, including shocks, and assess geometry-interface support, cache memory and runtime. The current experiment measures neither stability nor production speed.

Reproduction: docs/direct-derivatives.md; measurements/direct-derivatives/ contains all commands, raw rows, controls and source fingerprints. 50 configurations, 2644 rows, zero failed runs; 52.5 s of monitored numerical runs, one rank/thread on 8 logical processors, peak sampled RSS 21.0 MiB. The independent probes are additional small checks. All monitored cases stayed below 5 minutes and 4 GB.

The previous physical-polynomial reconstruction study is retained separately in measurements/physical-reconstruction/ and docs/physical-reconstruction.md, with its earlier PDF archived. Its pages are omitted here, including from the appendix, to keep this discussion focused.

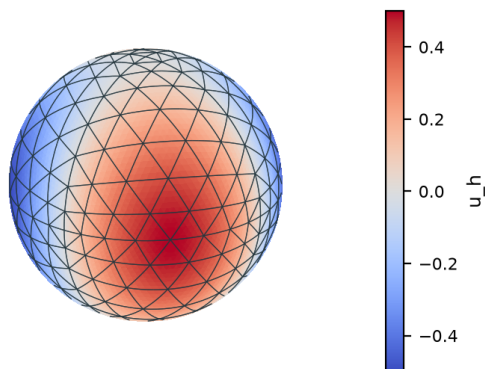
Solving a PDE on a spherical surface

Purpose: demonstrate and verify a solve on a curved two-dimensional mesh embedded in three-dimensional space. This is the surface itself, not a mesh filling the sphere. The stationary screened-Poisson problem works well as an inexpensive first example with a known solution.

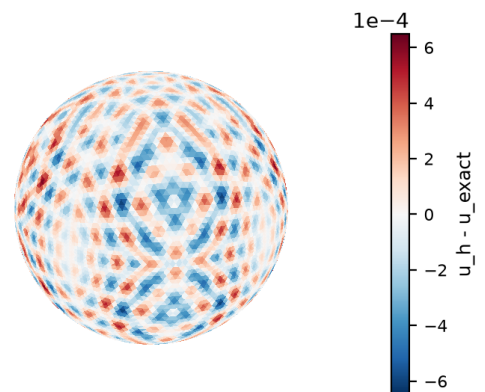
Problem: $-\Delta_\Gamma u + u = f$ on the unit sphere

Choose $u(x,y,z)=xy$ and $f=7xy$ on $\Gamma=\{x^2+y^2+z^2=1\}$, following the pinned MFEM Example 7. The spherical Laplacian satisfies $\Delta_\Gamma(xy)=-6xy$. The sphere has no boundary, so no boundary conditions are imposed; the $+u$ term removes the constant nullspace of the pure Laplacian.

Computed solution on the curved mesh



Signed error against radial exact extension



Computed quadratic solution and signed error on 512 curved triangles (refinement level 3, geometry degree 2). Left: original element edges on the visible side. Surface patches are subdivided only for rendering; colors average the three sampled vertex values per displayed patch. The coordinates come from the actual approximate surface, without projecting the rendered sphere to unit radius. The two panels use different color scales.

How surface geometry enters the solve

The element map F has two reference coordinates and three physical coordinates. Its Jacobian J is therefore 3×2 , not square. The metric G gives the tangential gradient and surface measure. We solve the weak problem below with continuous H^1 finite elements:

$$G = J^T J, \quad \nabla_{\Gamma_h} v = J G^{-1} \widehat{\nabla} \widehat{v}, \quad dS = \sqrt{\det G} d\xi$$

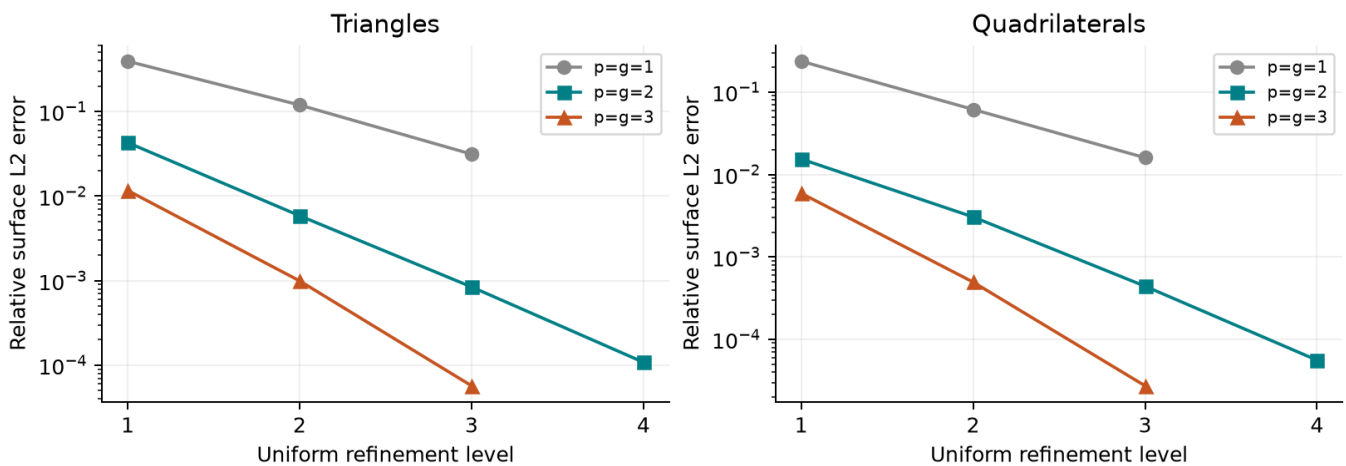
$$\int_{\Gamma_h} (\nabla_{\Gamma_h} u_h \cdot \nabla_{\Gamma_h} v_h + u_h v_h) dS = \int_{\Gamma_h} f^e v_h dS$$

MFEM supplies these surface transformations in its diffusion and mass integrators. High-order mesh nodes are snapped to the unit sphere after refinement. Between nodes, the polynomial surface Γ_h approximates rather than exactly equals Γ .

Scope and error reference

Away from the exact sphere, use the radial extensions $u^e(x,y,z)=xy/(x^2+y^2+z^2)$ and $f^e=7u^e$. Reported errors are integrated on Γ_h against u^e ; they include both discretization and geometry effects. This is a conforming surface-FEM result, not a DG/OFDG or surface-transport validation. Production OFDG remains unchanged.

Accuracy and geometry checks



p is solution degree; g is geometry degree. Triangles start from an octahedron, quads from a cube. Each refinement multiplies the element count by four. The p=g=2 series includes one additional refinement. These are not equal-DOF comparisons.

Elements / degree	Finest elements	Unknowns	Relative L2 error	L2 rate
Triangles / 1	512	258	3.136%	1.90
Triangles / 2	2048	4098	0.01084%	2.96
Triangles / 3	512	2306	0.005716%	4.12
Quads / 1	384	386	1.596%	1.93
Quads / 2	1536	6146	0.005587%	2.98
Quads / 3	384	3458	0.002732%	4.18

Rate = $\log_2(E_{\text{coarse}}/E_{\text{fine}})$ using absolute L2 errors on the final two levels. Observed rates approach p+1; tangential H1-seminorm rates are approximately p (retained in the data). This small manufactured example does not establish behavior for arbitrary surfaces or singular solutions.

Geometry matters: hold solution degree p=2 fixed

Level 3 / relative L2 error	Planar g=1	Curved g=2	Curved g=3
Triangles	1.253%	0.08434%	0.1143%
Quads	0.6944%	0.0441%	0.07463%

Curved geometry markedly improves this test over planar facets. Raising g from 2 to 3 reduces sphere-area error but increases the displayed solution error: geometry and solution-error contributions can interact or cancel. Better geometric accuracy alone is not a guarantee of monotonically smaller solution error.

Checks, resources and reproducibility

All 28 cases passed. A constant solution $u=f=1$ has relative L2 error at most $5e-14$. Raising assembly/evaluation quadrature by six changes L2 error by at most $1e-13$; the largest algebraic relative residual is $1.4e-12$. The closed-surface reaction balance $|(u_h - f^e)dS|$ is at most $2e-12$.

Sequential one-rank, one-thread execution: 5.61 s monitored compute, 23.1 MiB peak sampled RSS. Every check stayed below 5 minutes and 4 GB. Details and exact configurations: [docs/sphere-surface.md](#) and [measurements/sphere-surface/](#). Both triangle and quad solutions are saved as MFEM mesh/grid-function files for GLVis. Adapted from the installed [mfem/examples/ex7.cpp](#); no MFEM update was made.